



**HỌC VIỆN  
NGÂN HÀNG**  
BANKING ACADEMY OF VIETNAM

# **Web Design**

## **Lecture 3 - HTML (cont.)**



2.5. Adding images

2.6. Table markup

2.7. Forms

2.8. Embedded media

## 2.5. Adding images



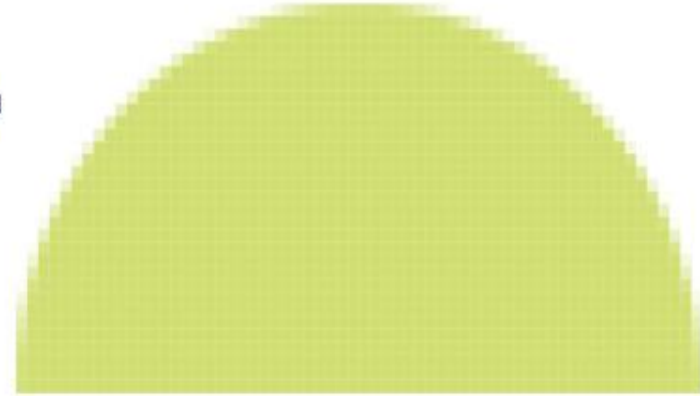
- Image formats
- The img element
- Responsive image markup



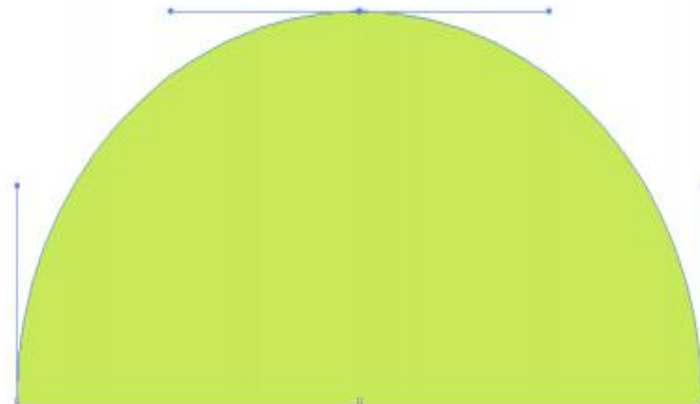
In general, images that are made up of a grid of colored pixels (called bitmapped or raster images) must be saved in the **PNG, JPEG, or GIF** file formats in order to be placed inline in the content.

For vector images, such as the kind of icons and illustrations you create with drawing tools such as Adobe Illustrator, we have the **SVG** format.

Bitmapped images are made up of a grid of colored pixels.



Vector images contain paths that are defined mathematically.





The **img** element tells the browser, “Place an image here.”

The **src** (source) attribute provides the location of the image file (its URL). The **alt** attribute provides alternative text that displays if the image is not available.

```
<p>This summer, try making pizza   
on your grill.</p>
```



This summer, try making pizza on your grill.



The value of the src attribute is the URL of the image file. In most cases, the images you use on your pages will reside on your own server, so you will use **relative URLs** to point to them.

### ***Performance Tip: Take Advantage of Caching***

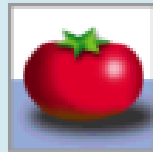
When a browser downloads an image, it stores the file in the **disk cache** (a space for temporarily storing files on the hard disk). That way, if it needs to redisplay the page, it can just pull up a local copy of the image without making a new server request.

---

---



The **width** and **height** attributes indicate the dimensions of the image in number of pixels



Fresh tomatoes.

`width="72" height="72"`  
(actual size of image)



Fresh tomatoes.

`width="144" height="72"`

## **Exercise 3-1 (10 mins). Adding and linking images**



In this exercise, you'll add images to pages and use them as links.

Textbook, page 158

---

---





After smartphones, tablets, “phablets,” and other devices hit the scene, large images that look great on a large screen were overkill on smaller screens:

- Slow down page display
- Data cost for downloading

Conversely, small images that download quickly may be blurry on large, high-resolution screens.

→ need a way to make whole web pages respond and adapt to various screen sizes → **“Responsive”**

---

---



In short, responsive images work this way: you provide multiple images, sized or cropped for different screen sizes, and the browser picks the most appropriate one based on what it knows about the current viewing environment.

- Providing extra-large images that look crisp on high-resolution screens
  - Providing a set of images of various dimensions for use on different screen sizes
  - Providing versions of the image with varying amount of detail based on the device size and orientation (known as the art direction use case)
  - Providing alternative image formats that store the same image at much smaller file sizes
- 
-



Do the exercise on page 156: Adding responsive images  
Test the result on various devices: mobile, tablet, laptop



How to use tables

Table structure

Spanning cells

Wrapping up tables

---

---



HTML tables were created for instances when you need to add tabular material (data arranged into rows and columns) to a web page.

Tables may be used to organize schedules, product comparisons, statistics, or other types of information

| PM  | 7:30                                       | 8:00                | 8:30                                             | 9:00                                                                                           | 9:30                                          | 10:00                      | 10:30 |
|-----|--------------------------------------------|---------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------|-------|
| ABC | <i>The Adventures of Ozzie and Harriet</i> | The Patty Duke Show | <i>Gidget</i>                                    | <i>The Big Valley</i>                                                                          |                                               | Amos Burke — Secret Agent* |       |
| CBS | Lost in Space                              |                     | <i>The Beverly Hillbillies</i><br>#8 25.9 rating | <i>Green Acres</i> #11<br>24.6 rating                                                          | <i>The Dick Van Dyke Show</i> #16 23.6 rating | <i>The Danny Kaye Show</i> |       |
| NBC | <i>The Virginian</i> #25 22.0 rating       |                     |                                                  | <i>Bob Hope Presents the Chrysler Theatre</i> /<br><i>Chrysler Presents a Bob Hope Special</i> |                                               | <i>I Spy</i>               |       |

wikipedia.org



All of the table's content goes into cells that are arranged into rows. Cells contain either header information (titles for the columns, such as “Calories”) or data, which may be any sort of content

| Menu item           | Calories | Fat |
|---------------------|----------|-----|
| Chicken noodle soup | 120      | 2   |
| Caesar salad        | 400      | 26  |



It is common to stack the th and td elements in order to make them easier to find in the source. This does not affect how the browser renders them.

| Menu item           | Calories | Fat (g) |
|---------------------|----------|---------|
| Chicken noodle soup | 120      | 2       |
| Caesar salad        | 400      | 26      |

```
<table>
  <tr>
    <th>Menu item</th>
    <th>Calories</th>
    <th>Fat (g)</th>
  </tr>
  <tr>
    <td>Chicken noodle soup</td>
    <td>120</td>
    <td>2</td>
  </tr>
  <tr>
    <td>Caesar salad</td>
    <td>400</td>
    <td>26</td>
  </tr>
</table>
```



One fundamental feature of table structure is cell spanning, which is the stretching of a cell to cover several rows or columns. Spanning cells allows you to create complex table structures, but it has the side effect of making the markup a little more difficult to keep track of.

You make a header or data cell span by adding the **colspan** or **rowspan** attributes

---

---





Column spans,  
created with the  
colspan attribute in  
the **td** or **th** element,  
stretch a cell to the  
right to span over the  
subsequent columns

```
<table>  
  <tr>  
    <th colspan="2">Fat</th>  
  </tr>  
  <tr>  
    <td>Saturated Fat (g)</td>  
    <td>Unsaturated Fat (g)</td>  
  </tr>  
</table>
```

| Fat               |                     |
|-------------------|---------------------|
| Saturated Fat (g) | Unsaturated Fat (g) |



Row spans, created with the rowspan attribute, work just like column spans, but they cause the cell to span downward over several rows

```
<table>
  <tr>
    <th rowspan="3">Serving Size</th>
    <td>Small (8oz.)</td>
  </tr>
  <tr>
    <td>Medium (16oz.)</td>
  </tr>
  <tr>
    <td>Large (24oz.)</td>
  </tr>
</table>
```

|                     |                |
|---------------------|----------------|
| <b>Serving Size</b> | Small (8oz.)   |
|                     | Medium (16oz.) |
|                     | Large (24oz.)  |



### Exercise 3-2. The table challenge

Page 174 - textbook

Your Content Here

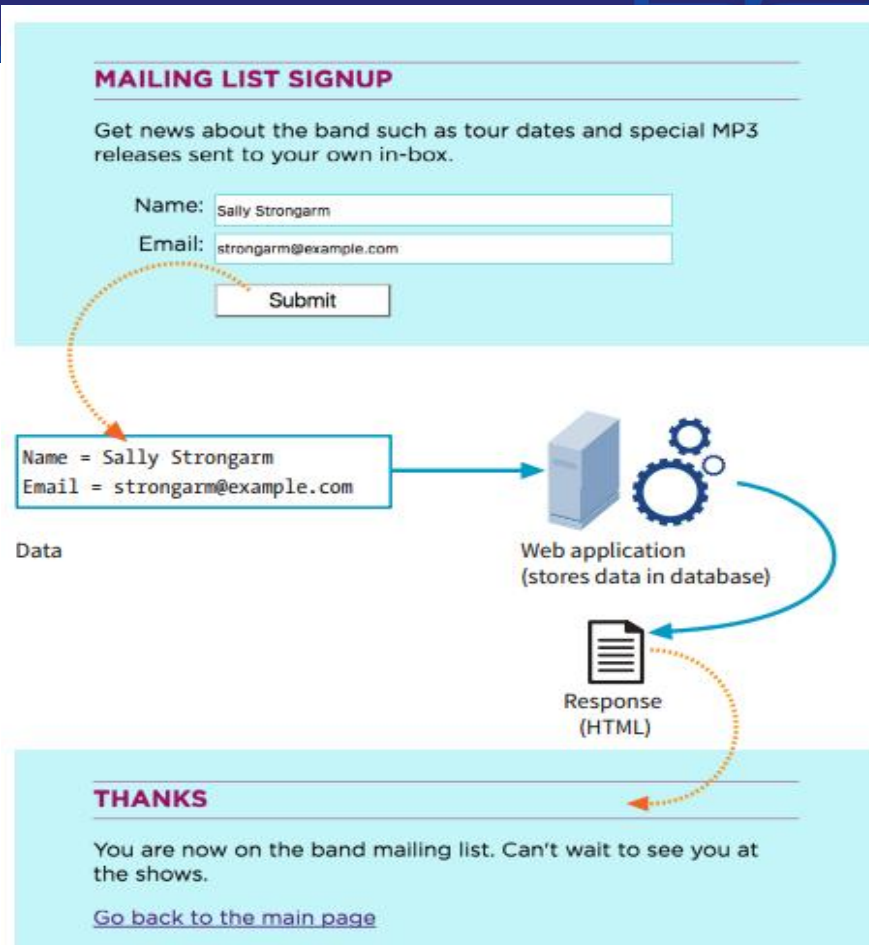
|         | A common header for two subheads |          | Header 3 |
|---------|----------------------------------|----------|----------|
|         | Header 1                         | Header 2 |          |
| Thing A | data A1                          | data A2  | data A3  |
| Thing B | data B1                          | data B2  | data B3  |
| Thing C | data C1                          | data C2  | data C3  |



- How forms work
- The form element
- Variables and content
- Form control
- Form accessibility features

There are two parts to a working form.

- The first part is the form that you **see on the page** itself that is created using HTML markup. Forms are made up of buttons, input fields, and drop-down menus (form controls) used to collect information from the user.
- The other component of a web form is an application or script **on the server** that processes the information collected by the form and returns an appropriate response. It's what makes the form work.





The form element is a container for all the content of the form, including some number of **form controls, such as text-entry fields and buttons**. It may also contain block elements (h1, p, and lists, for example). However, it ***may not contain another form element***.

```
<body>
  <h1>Mailing List Signup</h1>

  <form action="/mailinglist.php" method="POST">
    <fieldset>
      <legend>Join our email list</legend>
      <p>Get news about the band such as tour dates and special MP3
      releases sent to your own in-box.</p>
      <ol>
        <li><label for="firstlast">Name:</label>
          <input type="text" name="fullname" id="firstlast"></li>
        <li><label for="email">Email:</label>
          <input type="text" name="email" id="email"></li>
      </ol>
      <input type="submit" value="Submit">
    </fieldset>
  </form>

</body>
```



The **action** attribute provides the location (URL) of the application or script that will be used to process the form

```
<form  
action="/mailinglist.php"  
method="POST">...</form>
```

```
<body>  
  <h1>Mailing List Signup</h1>  
  
  <form action="/mailinglist.php" method="POST">  
    <fieldset>  
      <legend>Join our email list</legend>  
      <p>Get news about the band such as tour dates and special MP3  
releases sent to your own in-box.</p>  
      <ol>  
        <li><label for="firstlast">Name:</label>  
          <input type="text" name="fullname" id="firstlast"></li>  
        <li><label for="email">Email:</label>  
          <input type="text" name="email" id="email"></li>  
      </ol>  
      <input type="submit" value="Submit">  
    </fieldset>  
  </form>  
  
</body>
```



The **name** attribute provides the variable name for the control.

```
<textarea name="comment" rows="4" cols="45"  
placeholder="Leave us a comment."></textarea>
```

Naming Your Variables: be sure to name your variables simply and descriptively and to document them well

---

---





- Text-entry controls
- Specialized text-entry controls
- Submit and reset buttons
- Radio and checkbox buttons
- Pull-down and scrolling menus
- File selection and upload control
- Hidden controls
- Dates and times
- Numerical controls
- Color picker control

### Exercise 3-3. Starting the pizza order form

Page 191 - textbook

#### Black Goose Bistro | Pizza-on-Demand

Our 12" wood-fired pizzas are available for delivery. Build your custom pizza and we'll deliver it within an hour.

##### Your Information

Name:

Address:

Telephone Number:

Email:

Delivery instructions:

Limit characters and add placeholder text  
"No more than 400 characters long"

This form should be sent to  
<http://blackgoosebistro.com/pizza.php>  
via the POST method.

Name the text fields **customername**,  
**address**, **telephone**, **email**, and  
**instructions**, respectively.

##### Design Your Dream Pizza:

Pizza specs

Crust (Choose one):

- ☐ Classic white
- ☐ Multigrain
- ☐ Cheese-stuffed crust
- ☐ Gluten-free

Name the controls in this section **crust**,  
**toppings[]**, and **number**, respectively.

Note that the brackets ([ ]) after "toppings"  
are required in order for the script to  
process it correctly.

Toppings (Choose as many as you want):

- ☒ Red sauce
- ☐ White sauce
- ☐ Mozzarella Cheese
- ☐ Pepperoni
- ☐ Mushrooms
- ☐ Peppers
- ☐ Anchovies

Make sure "red sauce" is  
selected when the page loads.

Number

How many pizzas:

Pull down menu for  
ordering up to 6 pizzas.

Bring me a pizza!

Reset

Change the Submit button text.



IFrame

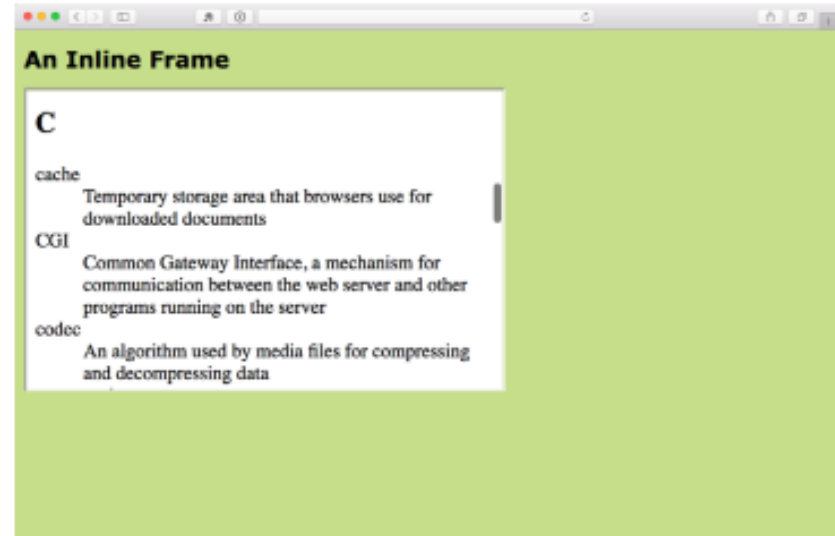
Video and audio

Canvas



The iframe (short for inline frame) element lets you embed a separate HTML document or other web resource in a document → Window in a window

```
<h1>An Inline Frame</h1>  
<iframe src="glossary.html" width="400" height="250" >  
  Read the <a href="glossary.html">glossary</a>.  
</iframe>
```





```
<video src="highlight_reel.mp4" width="640"  
height="480" poster="highlight_still.jpg" controls  
autoplay>
```

*Your browser does not support HTML5 video. Get  
the [MP4 video](highlight_reel.mp4)*

```
</video>
```

Browsers that **do not support video** display whatever content is provided within the video element. The good news is that the video and audio elements are **well supported in modern browsers**

---

---



The canvas element creates an area on a web page for drawing with a set of JavaScript functions for creating lines, shapes, fills, text, animations, and so on.

```
<canvas width="600" height="400"  
id="my_first_canvas">
```

***Your browser does not support HTML5 canvas. Try using Chrome, Firefox, Safari or MS Edge.***

```
</canvas>
```

---

---

## Exercise 3-4. Embed the Google maps to your page



<https://developers.google.com/maps/documentation/javascript/adding-a-google-map>

---

---